

Department of Electronics and Telecommunication Engineering
(NBA ACCREDITED)
Digital System Design Laboratory
Academic Year 2025-2026, SEM-5, Odd Semester

Course Code	EC24P
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Class	T.E – EXTC SEM-V
Division	B
Date of Performance	22-8-2025
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EXPERIMENT NO.6

Estimated time to complete this experiment: 02 hours.

Objective: To design and Implement 4 bit binary comparator.

CO to be achieved: CO1, CO2

Expected Outcome of Experiment:

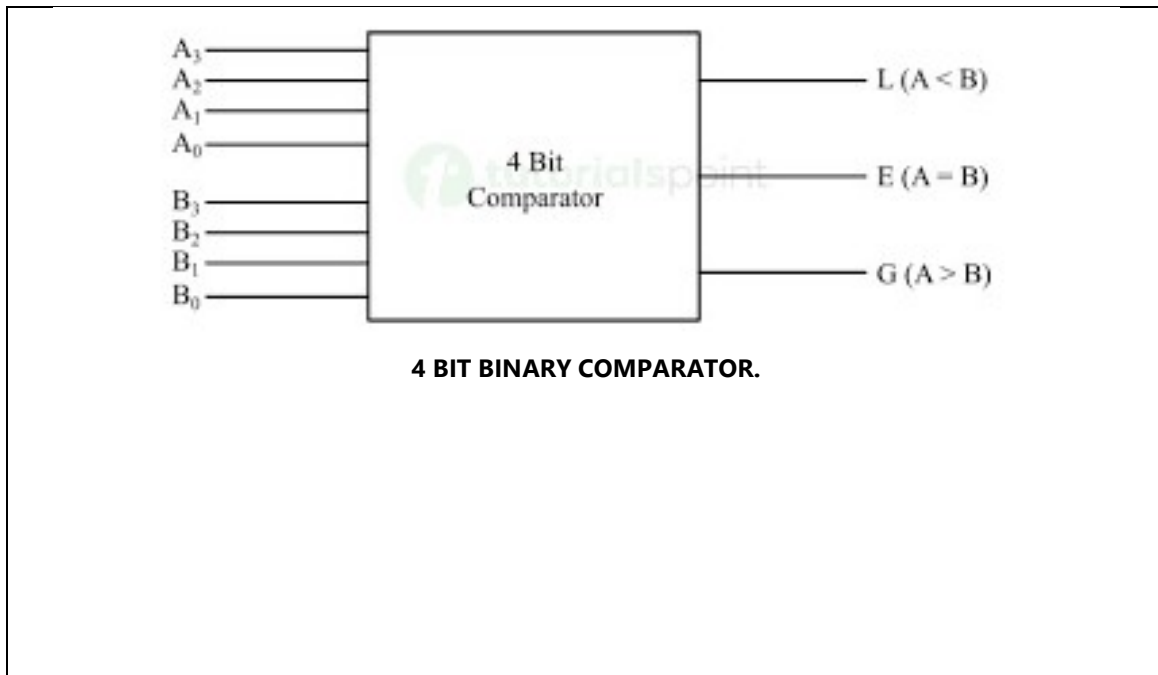
- Students will learn to design, simulate and synthesis the arithmetic circuit using VIVADO & VERILOG.
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Pre Lab/ Prior Concept: Designing of Binary adders.

Apparatus/TOOL:

- PC
 - VIVADO TOOL
 - SPARTABN-7 FPGA Board.
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CIRCUIT DIAGRAM.



VERILOG -CODE.

VERILOG CODE FOR 4 BIT BINARY COMPARATOR.

```
module mag_comp_4bit( input [3:0] A,input [3:0] B,output reg A_greater_B, output reg A_equal_B,
output reg A_less_B);
always @(*) begin
    if (A > B) begin
        A_greater_B = 1;
        A_equal_B = 0;
        A_less_B = 0;
    end
    else if (A == B) begin
        A_greater_B = 0;
        A_equal_B = 1;
        A_less_B = 0;
    end
    else begin
        A_greater_B = 0;
        A_equal_B = 0;
    end
end
```

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```
A_less_B = 1;
end
end
endmodule
```

TEST BENCH.

```
module tb_mag_comp_4bit();
reg [3:0] A, B;
wire A_greater_B, A_equal_B, A_less_B;
mag_comp_4bit uut ( .A(A), .B(B), .A_greater_B(A_greater_B), .A_equal_B(A_equal_B),
.A_less_B(A_less_B) );
initial begin
$monitor("Time=%0t | A=%b (%d) | B=%b (%d) | Greater=%b Equal=%b Less=%b",
$time, A, A, B, B, A_greater_B, A_equal_B, A_less_B);
A = 4'b0000; B = 4'b0000; #10;
A = 4'b1000; B = 4'b0100; #10;
A = 4'b0011; B = 4'b0101; #10;
A = 4'b1111; B = 4'b1111; #10;
A = 4'b0110; B = 4'b0111; #10;
A = 4'b1010; B = 4'b1010; #10;
$stop;
end
endmodule
```

VERILOG CODE USING CONDITIONAL INSTRUCTION.

```
module comp4bit_conditional ( input [3:0] A,input [3:0] B,
output A_greater_B,output A_equal_B, output A_less_B);

assign A_greater_B = (A > B) ? 1'b1 : 1'b0;
assign A_equal_B = (A == B) ? 1'b1 : 1'b0;
assign A_less_B = (A < B) ? 1'b1 : 1'b0;

endmodule

module tb_comp4bit_conditional();
reg [3:0] A, B;
wire A_greater_B, A_equal_B, A_less_B;
comp4bit_conditional uut ( .A(A), .B(B), .A_greater_B(A_greater_B), .A_equal_B(A_equal_B),
.A_less_B(A_less_B));

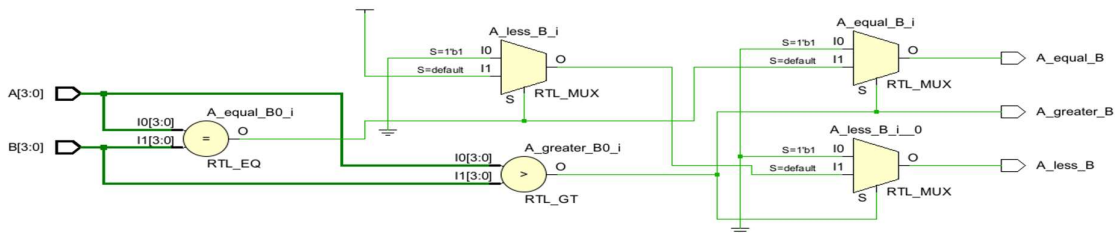
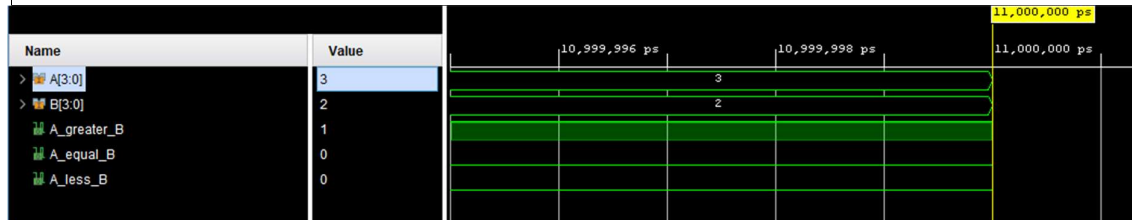
initial begin
$display("Time\tA\tB\tGreater\tEqual\tLess");
$monitor("%0t\t%b\t%b\t%b\t%b\t%b",
$time, A, B, A_greater_B, A_equal_B, A_less_B);
// Test cases
A = 4'b0000; B = 4'b0000; #10;
```

```
A = 4'b1000; B = 4'b0100; #10;
A = 4'b0011; B = 4'b0101; #10;
A = 4'b1111; B = 4'b1111; #10;
A = 4'b0110; B = 4'b0111; #10;
A = 4'b1010; B = 4'b1010; #10;
A = 4'b0001; B = 4'b1000; #10;
$stop; // End simulation
```

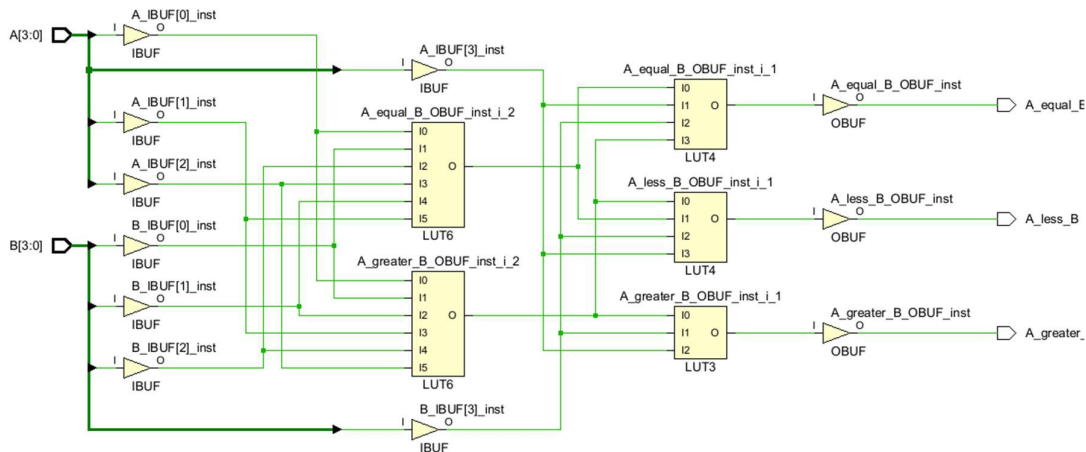
end

endmodule

SIMULATION RESULTS.



SYNTHESIS MODEL.



CONCLUSION.

The 4-bit comparator was successfully designed and implemented using Verilog on the FPGA board. The circuit correctly compared two 4-bit binary inputs and generated outputs indicating whether the numbers were equal, greater, or less. The results matched the expected truth table, verifying the correct working of the comparator.

POST LAB QUESTION.

1. Design the 4 bit magnitude comparator using Gate Level Modelling.

```
module comparator_4bit(A, B, EQ, GT, LT);  
    input [3:0] A, B;  
    output EQ, GT, LT;  
  
    wire e3, e2, e1, e0;  
    wire g3, g2, g1, g0;  
    wire l3, l2, l1, l0;  
  
    // Equality check (XNOR for each bit)  
    xnor(e3, A[3], B[3]);  
    xnor(e2, A[2], B[2]);  
    xnor(e1, A[1], B[1]);  
    xnor(e0, A[0], B[0]);  
  
    and(EQ, e3, e2, e1, e0);  
  
    // Greater than logic  
    and(g3, A[3], ~B[3]);  
    and(g2, e3, A[2], ~B[2]);  
    and(g1, e3, e2, A[1], ~B[1]);  
    and(g0, e3, e2, e1, A[0], ~B[0]);  
    or(GT, g3, g2, g1, g0);  
  
    // Less than logic  
    and(l3, ~A[3], B[3]);  
    and(l2, e3, ~A[2], B[2]);  
    and(l1, e3, e2, ~A[1], B[1]);  
    and(l0, e3, e2, e1, ~A[0], B[0]);  
    or(LT, l3, l2, l1, l0);  
endmodule
```

ASSESSMENT RECORD.

Practical Performance [Model/Code, Results] (5 Marks)	Presentation/ Learning Outcome [Observation, Conclusion] (5 Marks)	Total (10 Marks)	Sign